PixDiff: Multiresolution Diffusion Network With Pixelization for Hyperspectral Anomaly Detection

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***Abstract*—Hyperspectral anomaly detection (HAD) is a crucial area in hyperspectral image (HSI) processing, aiming to identify unknown anomalous objects. Since anomalies are usually more difficult for networks to learn compared with backgrounds, a popular deep learning-based paradigm is to distinguish anomalies from the background through reconstruction errors. However, anomaly regions in the reconstruction objective may still guide the network during training. In addition, complex backgrounds in HSI that are difficult to reconstruct may be challenging to differentiate from anomalies. Due to the unique spatial characteristics of anomalies and complex backgrounds, structural spatial transformations can be applied to create distinguishable differences between them. In this article, a multiresolution diffusion network based on pixelization called PixDiff is proposed to reduce the visibility of anomalies and differentiate them from complex backgrounds. In PixDiff, multiscale pixelization is innovatively introduced into HAD to preserve coarse background structures while disrupting anomaly information. Then, a structural isolation module (SIM) is proposed to distinguish difficult-to-reconstruct complex backgrounds from potential anomalies through interresolution differences. Furthermore, a multiresolution adaptive weighted reconstruction loss is proposed to further increase the distinction between anomalies and complex backgrounds. Experimental results on five HAD datasets demonstrate that the proposed PixDiff achieves superior detection performance compared with existing state-of-the-art methods.**

***Index Terms*—Anomaly detection, diffusion model, hyperspectral image (HSI), multiresolution network, pixelization.**

I. Introduction

HYPERSPECTRAL images (HSIs), unlike conventional RGB images, usually encompass hundreds or even thousands of spectral bands [1]. This enables HSIs to incorporate rich information about land cover features. Consequently, HSIs have been widely applied in various image processing tasks of remote sensing research, such as land cover classification [2], [3], [4], spectral unmixing [5], [6], change detection [7], [8], and anomaly detection [9], [10], [11]. Hyperspectral anomaly detection (HAD) specifically aims to identify unknown anomalous regions or objects within hyperspectral scenes. These anomalies usually have spectral characteristics significantly different from the background and surrounding features, such as aircraft and ships in military applications or polluted areas in environmental applications. Generally, anomaly samples constitute a very small proportion of the data, and their prior spectral information is difficult to obtain. Therefore, HAD methods based on the machine learning usually adopt unsupervised or weakly supervised strategies [12], [13].

Existing HAD methods can be broadly categorized into statistics-based methods [9], [14], [15], [16], [17], [18], [19], [20], representation-based methods [21], [22], [23], [24], [25], [26], [27], [28] and deep learning-based methods [12], [29], [30], [31], [32], [33]. Statistics-based methods mostly assume that the HSI background follows specific distributions, such as the Gaussian distribution, while anomalies deviate from these distributions, thereby enabling the anomaly detection by measuring the distance between samples and the assumed distribution. However, these background distribution assumptions may not be applicable to all scenes, particularly when dealing with backgrounds composed of complex land covers. Among representation-based methods, collaborative representation [21], [34], [35], [36], matrix decomposition [22], [23], [24], [25], and tensor representation [23], [27], [28], [37], [38], [39] are commonly used approaches. Collaborative representation-based methods generally assume that background pixels can be linearly represented by their surrounding pixels, while anomalies cannot. Matrix decomposition and tensor representation-based methods usually view HSI as a combination of low-rank background and sparse anomalies. Therefore, the background can be reconstructed through dictionaries, and anomalies that cannot be represented will be excluded. These representation-based methods often require a series of parameters for each HSI to construct appropriate surrounding regions or background dictionaries, which increases the complexity and uncertainty of the methods.

With the rapid development of deep learning, numerous HAD methods based on deep neural networks (DNNs) have emerged in recent years, aiming to model HSIs and detect anomalies through the powerful representation capabilities of DNNs. Since anomalies usually constitute a small proportion of the data and differ significantly from the background, it can be assumed that anomalies are more difficult for networks to learn than the background. Based on this assumption, a popular strategy is to reconstruct HSI through DNNs and identify anomalies by measuring the difference between the reconstructed result and the original image. However, due

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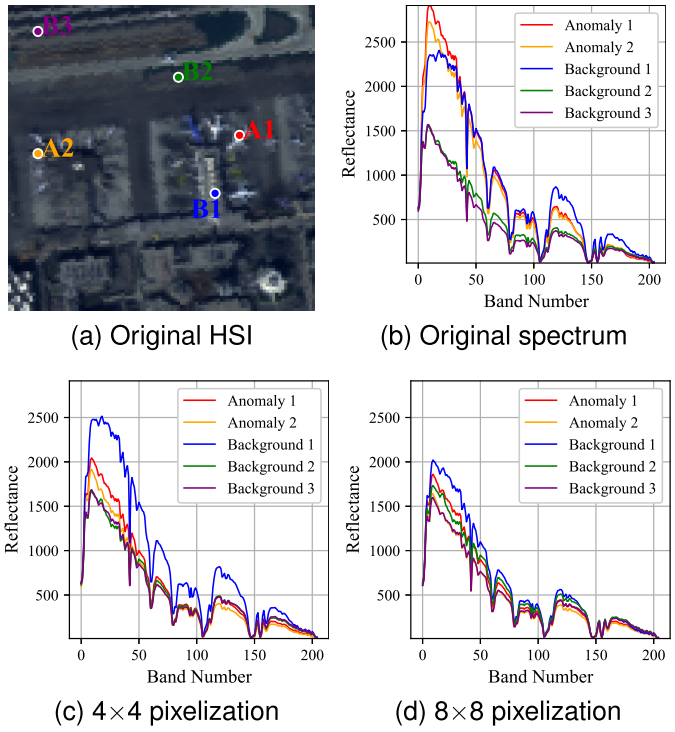


Fig. 1. Visualization of the anomaly and background spectral reflectance curves after pixelation. (a) Location of the selected pixel in the original HSI. (b) Spectral reflectance curve of the original pixel. (c) Spectral reflectance curve of the pixel after 4 × 4 pixelization. (d) Spectral reflectance curve of the pixel after 8 × 8 pixelization.

to DNN's powerful learning ability, anomalies may still be gradually reconstructed by training. Therefore, many methods have been introduced to address this issue, such as adaptive weighted loss functions [12], separation training [31], and gating mechanisms [30]. They identify potential anomalies at the early training steps and continuously update suspicious regions, thereby mitigating the effect of anomalies on network learning. Some recent methods introduce the denoising diffusion probabilistic model (DDPM). By adding noise similar to HSI distribution, the network can gradually suppress the background or restore anomalies to the background during the denoising process, further improving the performance of DNN [40], [41].

In summary, these reconstruction-based DNN methods are mostly developed around this principle: reconstructing the background while excluding the learning of anomalies. However, these methods may face the following issues.

1) Anomalies may still be visible in learning. The structure of anomalies in the original HSI remains intact and still guides the network as the reconstruction objective.
2) Complex backgrounds are difficult to distinguish from anomalies. There may be complex backgrounds in HSI that are difficult to reconstruct, which may lead the network to suppress them along with anomalies.
3) The diversified modeling capabilities of DDPM are not fully utilized. In the process of learning a single-resolution image, the network may not necessarily extract the representative background features.

To address the above issues, we propose a pixelization-based multiresolution diffusion network called PixDiff. Through observation, we find that although both complex backgrounds and anomalies are difficult to reconstruct in the network, their spatial patterns may differ. Complex backgrounds that are difficult to reconstruct, such as edges and buildings, mostly have larger and more continuous ranges in images and rarely exhibit obvious target morphology. Therefore, we can apply structural spatial transformations to the image to reduce the visibility of anomalies, creating distinguishable differences between anomalies and complex backgrounds. Based on this motivation, we innovatively introduce pixelization into HAD to extract the overall structural features of land covers while disrupting anomalies. As shown in Fig. 1, we analyzed the effect of pixelization on the spectral reflection curves of anomalies and backgrounds. In the original HSI, compared with the basic backgrounds B2 and B3, anomalies A1 and A2, and complex background B1 all have large deviations, which may cause confusion. After pixelization, complex background B1 is more difficult to converge toward the basic background than anomalies due to its surrounding spatial structural characteristics. Thus, we can assume that regions that are difficult to reconstruct in pixelized images are more likely to belong to complex backgrounds rather than anomalies. Based on the above analysis, we designed a multiresolution reconstruction diffusion network, utilizing the structural differences of images at different resolutions to distinguish complex backgrounds and anomaly regions.

Specifically, we introduce pixelization into DDPM to learn the reconstruction process of varying resolutions from noise. A shared encoder captures common features across different resolutions, followed by three decoder branches dedicated to reconstructing multiresolution images. This architecture enables the network to focus on common background structures, reduce the visibility of anomalies in the network, while preserving the background details of the original image. To distinguish between complex backgrounds and anomalies in reconstructed images, we designed a structural isolation module (SIM) based on interresolution differences. SIM analyzes the reconstruction errors among resolutions and then sifts potential complex backgrounds from anomalies. Since pixelization may cause position deviations from the original image, we designed an intrablock expanding method to calculate the reconstruction error of the pixelized image, enabling fuzzy alignment between different resolutions. Furthermore, we designed a multiresolution adaptive weighted reconstruction loss. It guides different resolution branches to handle anomalies and backgrounds differently based on the obtained information from SIM. In complex backgrounds, it makes the network focus more on learning detailed information at original resolution, while in anomaly regions, the network learns more about the information in pixelized resolution that is assimilated by the surrounding background. Therefore, the distinction between anomalies and complex backgrounds further increases in the network, while achieving fusion of multiresolution information within the network. Experimental results demonstrate that the proposed PixDiff achieves superior detection performance compared with state-of-the-art methods on five real-world datasets.

The main contribution of our work can be summarized as follows.

1) A pixelization-based multiresolution diffusion network is proposed. Through pixelization, original anomaly

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structures are disrupted or assimilated by background pixels, guiding the network to ignore learning anomalies. Combined with the diffusion model, the network reconstructs pixelized images of different resolutions from Gaussian noise, enhancing the learning of representative background structure.

2) A complex background removal module called SIM is proposed. SIM analyzes the spatial structural characteristics of difficult-to-reconstruct regions of different pixelized resolutions, effectively distinguishing between complex backgrounds and anomaly pixels.
3) A multiresolution adaptive weighted reconstruction loss is proposed. It adaptively guides the enhancement or suppression of complex backgrounds and anomalies for reconstruction branches of different resolutions, achieving targeted fusion of information from various pixelization resolutions while further increasing the distinction between anomalies and complex backgrounds.
4) Extensive comparative and ablation experiments on five real-world datasets demonstrate that the proposed PixDiff achieves superior detection performance compared with existing state-of-the-art methods.

The remainder of this article is organized as follows. In Section II, we briefly review the previous related work on HAD and DDPM. In Section III, we detail the proposed PixDiff. In Section IV, we present experimental results and analysis to demonstrate the performance of the proposed PixDiff. Finally, we summarize the conclusion in Section V.

II. Related Works

A. Statistics-Based and Representation-Based Methods

Statistical-based methods and representation-based methods are two important branches in HAD research. Early HAD research mostly focused on statistical-based methods, among which the Reed–Xiaoli (RX) algorithm [9] is the most representative. The RX algorithm is based on the generalized likelihood ratio test (GLRT), assuming that the background of HSIs follows Gaussian distributions, and then uses the Mahalanobis distance to measure the difference between pixels and the background distribution. Subsequently, many RX-based improvement methods were proposed, such as local RX [14] and weighted RX (WRX) [15]. Some works mapped the original HSIs to other feature spaces to better utilize the RX detector, such as kernel RX (KRX) [16] and fractional Fourier entropy RX (FrFE-RX) [19]. Liu et al. [20] further applied GLRT to multipixel anomaly detection and proposed its two-step modification, reducing the computational complexity.

Representation learning is also widely applied in HAD, including collaborative representation, matrix decomposition, and tensor representation. Collaborative representation detection (CRD) [21] assumes that each background pixel can be represented by a linear combination of its neighborhood pixels, while anomalies cannot. Based on this assumption, several improved CRDs have been proposed, such as recursive CRD [34], relaxed CRD [35], and self-paced probabilistic CRD [42]. Matrix decomposition-based methods consider that the HSI matrix can be decomposed into low-rank backgrounds, sparse anomalies, and noise. Therefore, a low-rank and sparse (LRASR) detector [22] and a Mahalanobis distance method based on LRASR decomposition [23] were proposed. Zhao et al. [43] estimate the usage frequency of each dictionary atom in reconstruction, thereby evaluating the sparsity of pixels. Huyan et al. [24] constructed potential anomaly dictionaries to utilize the prior information of possible anomalies in HSIs. To further separate sparse anomalies from complex mixed noise, Guo et al. [25] proposed an antinoise hierarchical mutual-incoherence-induced discriminative learning (AHMID) HAD method. To preserve the 3-D tensor structure of HSI, many tensor-based methods have been proposed in recent years [26], [27], [28]. To utilize the piecewise smoothness information in the HSI background, Sun et al. [38] introduced tensor average rank and total variation (TV) regularization to reconstruct the background. Feng et al. [39] introduced TV regularization into tensor ring decomposition, so that the low-rank characteristics and piecewise smoothness of the background in spatial and spectral dimensions can be utilized simultaneously.

These statistics and representation learning-based methods are able to perform well, but there are still some shortcomings. Specifically, statistical-based methods are built on distributional assumptions and may not perform well on complex backgrounds. Representation-based methods may be sensitive to parameters and require manual adjustment for each image. These limitations are likely to restrict their applications in practical scenes.

B. Deep Learning-Based Methods

In recent years, reconstruction-based methods have become popular in deep learning-based HAD. Most of them assume that anomalies are more difficult to reconstruct by networks, so DNNs can be used to learn HSI reconstruction, and then distinguish anomaly samples based on reconstruction errors [44], [45], [46], [47]. A common framework is to treat each pixel (or patch containing the neighborhood information) as a sample for reconstruction, thereby calculating the anomaly score for each pixel [48]. Fan et al. [49] designed a graph regularization term based on superpixel to maintain the geometric structure of the background. Wang et al. [50], [51] adopted a dual-window blind-block structure, using surrounding pixels to reconstruct the center pixel, thus suppressing the visibility of anomalies in the network. To further ignore anomaly reconstruction, Lian et al. [30] designed an adaptive gating unit applied to the Transformer, using content similarity to process potential anomalies and backgrounds separately. Due to the possible existence of multiple land covers in the background, a novel deep feature aggregation network (DFAN) [52] was proposed to represent multiple background patterns through adaptive aggregation.

Another approach is to treat the entire image as a sample for processing, with the network directly outputting the reconstructed image or detection map. Wang et al. [12] proposed Auto-AD, using a fully convolutional autoencoder for automated HAD on the entire image without any preprocessing or postprocessing. In addition, Auto-AD designed an adaptive weighted loss, which makes the network ignore potential anomalies by assigning lower weights to regions with larger reconstruction errors. Liu et al. [31] further applied a separate training algorithm, optimizing potential anomalies and backgrounds separately through soft thresholding, and combined multiscale information to suppress the visibility and reconstruction of anomalies in the network. Yang et al. [53]

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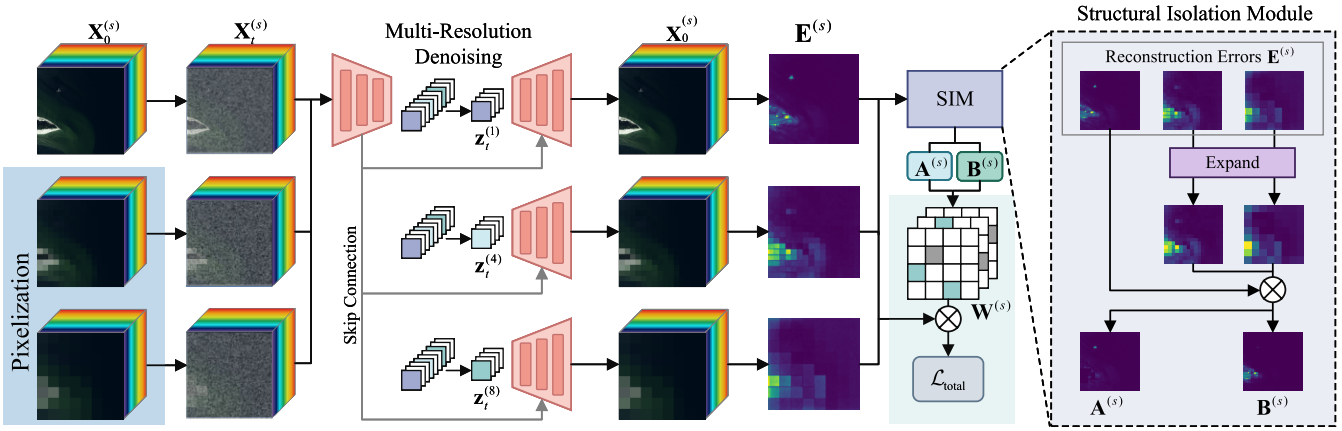


Fig. 2. Flowchart of the proposed PixDiff. First, forward diffusion is performed on the original HSI and pixelized images, followed by reconstruction of multiresolution images by the denoising network. The SIM module analyzes interresolution reconstruction differences and separates potential complex backgrounds and anomalies. Based on the separated maps, a multiresolution adaptive weighted loss is calculated to guide the network in discriminating between anomalies and complex background at each resolution.

adopted the blind-spot network and combined it with dynamic fusion to better extract the multiscale information of HSI. To utilize Mamba's capability in modeling long sequences, a Multiscale Mamba Reconstruction network was proposed, extracting and enhancing background information to suppress anomaly reconstruction [54]. These entire image-based methods can better consider the spatial relationships between pixels and maintain the spatial continuity of backgrounds.

These reconstruction-based methods focus on suppressing the learning of anomalies while effectively reconstructing the background. However, when confronted with complex backgrounds that may be confused with anomalies, the network may fail to distinguish them well, resulting in either simultaneous enhancement or suppression of both the complex background and anomalies. This limitation potentially constrains the detection performance of the model.

C. Denoising Diffusion Probabilistic Models

The DDPM is a generative model based on Markov chains that consists of a series of forward and reverse processes [55]. In the forward process, the Gaussian noise is gradually added to the data, while the network learns to progressively remove noise and restore the original image through the reverse process. Eventually, the network acquires the capability to recover meaningful data distributions from pure noise. Due to DDPM's excellent controllability, stability, and ability to generate rich and high-quality detailed features, it has been widely applied in various fields such as image generation [56] and image restoration [57].

In recent years, DDPM has been introduced into HAD, utilizing its powerful detail representation capabilities to better characterize backgrounds. Ma et al. [40] proposed BSDM, which first introduced diffusion models into HAD. BSDM treats the HSI background as removable noise and performs background suppression by learning the denoising process. Chen et al. [41] designed a spectral diffusion model that views the diffusion process as a gradual mixing of central pixels with surroundings. However, due to DDPM's powerful detail representation capability for both backgrounds and anomalies, it may treat background and anomaly features equivalently during reconstruction or suppression, potentially affecting the detection performance.

III. Proposed Method

A. Overview Framework

The proposed PixDiff mainly consists of three parts: pixelization processing, multiresolution diffusion, and structural isolation. Since the original HSI can be regarded as an image with the pixelized block size of 1×1, let the original HSI be defined as $\mathbf{X}^{(1)} \in \mathbb{R}^{H \times W \times B}$, where H, W, and B represent height, width, and number of bands, respectively. PixDiff first generates two pixelized images $\mathbf{X}^{(4)}$ and $\mathbf{X}^{(8)}$ of different resolutions, obtaining the overall network input $\mathcal{X} = \{\mathbf{X}^{(1)}, \mathbf{X}^{(4)}, \mathbf{X}^{(8)}\}$. Subsequently, the multiresolution diffusion network learns to reconstruct images at three different resolutions through noising and denoising processes, obtaining reconstructed images $\hat{\mathcal{X}} = \{\hat{\mathbf{X}}^{(1)}, \hat{\mathbf{X}}^{(4)}, \hat{\mathbf{X}}^{(8)}\}$. Reconstruction differences across multiresolution images are analyzed by the SIM, which generates complex background maps and potential anomaly maps. Based on the identified complex backgrounds and potential anomalies, an adaptive weight $\mathcal{W} = \{\mathbf{W}^{(1)}, \mathbf{W}^{(4)}, \mathbf{W}^{(8)}\}$ is constructed during training to guide the reconstruction process for images at different resolutions. During the inference phase, the potential anomaly map extracted by SIM serves as the detection result by PixDiff. The structure of the proposed PixDiff is shown in Fig. 2.

B. Pixelization-Based Multiresolution Diffusion Network

1) Pixelization Processing: Given an original HSI $\mathbf{X}^{(1)}$, we generate two pixelized images of $\mathbf{X}^{(1)}$ at different resolutions. Within each pixelized block, we replace the pixel values with the mean of the corresponding region in the original image, thereby preserving the general structure of the background while disrupting anomalies. Specifically, let s be the side length of a pixelized block. For any pixel position (i, j), we first determine which pixelized block it belongs to using $\lfloor \frac{i}{s} \rfloor$ and $\lfloor \frac{j}{s} \rfloor$, where $\lfloor \cdot \rfloor$ represents the floor function that maps the pixel to its block index. Then, we average all $s \times s$ pixels

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within that block. The pixelized image $\mathbf{X}_{i,j}^{(s)}$ can be expressed as follows:

$$\mathbf{X}_{i,j}^{(s)} = \frac{1}{s^2}\sum_{p=0}^{s-1}\sum_{q=0}^{s-1}\mathbf{X}^{(1)}_{\lfloor \frac{i}{s}\rfloor\cdot s+p,\lfloor \frac{j}{s}\rfloor\cdot s+q}. \tag{1}$$

A single size for pixelized blocks may not be able to cover anomalies of different sizes, while excessively large pixelized blocks might compromise too much of the background structure. Therefore, in this work, we select $\mathbf{X}^{(4)}$ and $\mathbf{X}^{(8)}$ as the medium-resolution and low-resolution images. This enables the handling of unknown anomalies of different sizes and generates differences between various resolution levels that are worth capturing.

2) Multiresolution Diffusion Network: PixDiff learns the reconstruction of HSI through the forward and reverse process based on DDPM [55]. For every pixelization scale s, the forward diffusion process of DDPM can be expressed as follows:

$$q\left(\mathbf{X}_t^{(s)}|\mathbf{X}_{t-1}^{(s)}\right) = \mathcal{N}\left(\mathbf{X}_t^{(s)};\ \sqrt{1-\beta_t}\mathbf{X}_{t-1}^{(s)},\beta_t\mathbf{I}\right) \tag{2}$$

where $\mathbf{I}$ is the identity matrix and β_t is the predefined noise scheduling parameter. In this work, we use linear scheduling to generate β_t. Given the maximum value $\beta_{\max}$ and minimum value $\beta_{\min}$, and the total number of steps T, the parameter β_t at time t can be calculated as follows:

$$\beta_t = \beta_{\min} + \frac{t}{T}\left(\beta_{\max}-\beta_{\min}\right). \tag{3}$$

We adopted the parameterization process from DDPM, which enables sampling at any time step. For each time step, $\mathbf{X}_t^{(s)}$ can be represented by the initial image $\mathbf{X}^{(s)}$ as follows:

$$\mathbf{X}_t^{(s)} = \sqrt{\bar{\alpha}_t}\mathbf{X}^{(s)} + \sqrt{(1-\bar{\alpha}_t)}\epsilon, \quad \epsilon\sim\mathcal{N}(0,\mathbf{I}) \tag{4}$$

where $\bar{\alpha}_t$ can be calculated from β_t as follows:

$$\alpha_t = 1-\beta_t \tag{5}$$

$$\bar{\alpha}_t = \prod_{s=0}^{t}\alpha_s. \tag{6}$$

After the forward process, the multiresolution noised images at time t can be represented as $\mathcal{X}_t = \{\mathbf{X}_t^{(1)},\mathbf{X}_t^{(4)},\mathbf{X}_t^{(8)}\}$. During training, we use the multiresolution noisy images from the same time step as network inputs to compare their reconstruction errors and extract the complex backgrounds. During inference, three noise samples with the same size as $\mathbf{X}^{(1)}$ are sampled as $\mathcal{X}_t$.

The denoising network consists of a feature extraction encoder and a multiresolution reconstruction decoder. The network is mainly composed of convolutional layers and self-attention modules, combining the local and global receptive fields. Following the U-Net structure, several residual connections are used between the encoder and decoder. After the encoder extracts the common features of multiresolution, we split them into the latent features $\mathbf{z}_t^{(s)}$ corresponding to each resolution. Then, we use three decoder branches to learn the reconstruction of details at different resolutions. Through the denoising network, multiresolution reconstructed images $\hat{\mathcal{X}} = \{\hat{\mathbf{X}}^{(1)},\hat{\mathbf{X}}^{(4)},\hat{\mathbf{X}}^{(8)}\}$ can be obtained. The corresponding reconstruction errors $\mathcal{E} = \{\mathbf{E}^{(1)},\mathbf{E}^{(4)},\mathbf{E}^{(8)}\}$ can be calculated as follows:

$$\mathbf{E}_{i,j}^{(s)} = \|\mathbf{X}_{i,j}^{(s)} - \hat{\mathbf{X}}_{i,j}^{(s)}\|_2. \tag{7}$$

3) Structural Isolation Module: After obtaining the reconstruction errors at different resolutions, we extract potential complex backgrounds through their differences. Due to pixel position deviations after pixelization, for the two pixelized images, we first expand the reconstruction errors within the pixelized blocks. For side length s of the pixelized block, the processed reconstruction error map $\mathbf{E}^{(s)*}$ can be expressed as follows:

$$\mathbf{E}_{i,j}^{(s)*} = \max_{0\le p<s,0\le q<s}\mathbf{E}^{(s)}_{\lfloor \frac{i}{s}\rfloor\cdot s+p,\lfloor \frac{j}{s}\rfloor\cdot s+q}. \tag{8}$$

In pixelized images, content that is difficult to reconstruct is more likely to be the complex background. Based on this assumption, potential complex backgrounds in the original image can be extracted as follows:

$$\mathbf{B}^{(1)} = \lambda\mathbf{E}^{(1)}\odot\mathbf{E}^{(4)*} + (1-\lambda)\,\mathbf{E}^{(1)}\odot\mathbf{E}^{(8)*} \tag{9}$$

where λ is a balance parameter, $\odot$ represents the Hadamard product. Then, the anomaly map $\mathbf{A}$ separated from the complex background can be calculated as follows:

$$\mathbf{A}^{(1)} = \mathbf{E}^{(1)} - \mathbf{B}^{(1)}. \tag{10}$$

C. Multiresolution Adaptive Weighted Reconstruction Loss

To suppress reconstruction of anomalies, some existing methods, such as Auto-AD [12], design adaptive weights to make the network ignore learning in high reconstruction error regions. However, these methods may suppress complex backgrounds together with anomalies, further increasing the reconstruction error of complex backgrounds that are difficult to reconstruct. Based on the complex background and anomalies extracted by SIM, we design an adaptive weighted reconstruction loss for the multiresolution network, thereby increasing the distinction between anomalies and complex backgrounds.

For the original resolution, we aim to enhance the reconstruction of complex backgrounds while ignoring anomalies. Therefore, based on the all-ones matrix $\mathbf{J}\in\mathbb{R}^{H\times W\times 1}$, we increase the weights of potential complex background regions according to $\mathbf{B}^{(1)}$ and decrease the weights of potential anomaly regions according to $\mathbf{A}^{(1)}$. In summary, the weight for the original resolution is designed as follows:

$$\mathbf{W}^{(1)} = \mathbf{J} + \mathbf{B}^{(1)} - \mathbf{A}^{(1)}. \tag{11}$$

For pixelized images, we aim to make the network ignore complex background details, thereby further enhancing the differences in reconstruction of complex backgrounds between different resolutions. To obtain complex background regions in pixelized images, we need to exclude potential anomaly areas from the reconstruction errors of pixelized images. First, the separated anomalies are expanded within the entire pixelized block as follows:

$$\mathbf{A}_{i,j}^{(s)} = \max_{0\le p<s,0\le q<s}\mathbf{A}^{(1)}_{\lfloor \frac{i}{s}\rfloor\cdot s+p,\lfloor \frac{j}{s}\rfloor\cdot s+q}. \tag{12}$$

Then, the potential complex backgrounds of the pixelized image can be calculated as follows:

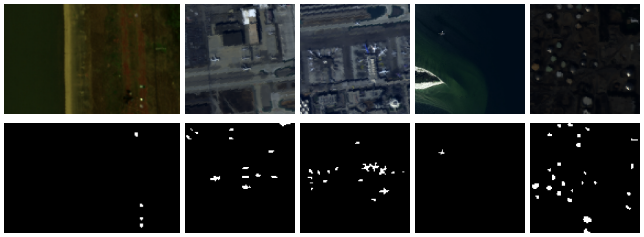
$$\mathbf{B}^{(s)} = \mathbf{E}^{(s)} - \mathbf{A}^{(s)}, \quad s\in\{4,8\}. \tag{13}$$

Contrary to the original resolution, we assign lower weights to potential complex background regions, thereby making the pixelized resolution branch ignore their reconstruction.

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TABLE I
AUROC SCORES FOR DIFFERENT METHODS ON THE FIVE DATASETS

Dataset	RX	CRD	LRASR	2S-GLRT	PTA	LREN	Auto-AD	BSDM	BockNet	GT-HAD	DFAN	MSNet	PixDiff
WHU-Hi-River	0.9988	0.9755	0.9130	0.9915	0.8828	0.9844	0.9987	0.9988	0.9994	0.9981	0.9987	**0.9995**	**0.9995**
Los Angeles-I	0.8978	0.8403	0.6483	0.8157	0.8203	0.8482	0.8831	0.8978	0.9367	0.8813	0.8988	0.9312	**0.9418**
Los Angeles-II	0.9416	0.6517	0.5516	0.6914	0.7734	0.9166	0.9163	0.9416	0.9518	0.9420	0.9431	0.9491	**0.9672**
Cat Island	0.9763	0.9943	0.8932	0.9959	0.9801	0.8951	0.9707	0.9763	0.9872	0.9986	0.9778	0.9988	**0.9998**
Urban	0.9802	0.7780	0.6403	0.7696	0.7674	0.9725	0.9796	0.9802	0.9684	0.9745	0.9805	0.9804	**0.9891**
Mean	0.9589	0.8480	0.7293	0.8528	0.8448	0.9234	0.9497	0.9589	0.9687	0.9589	0.9598	0.9718	**0.9795**



(a) (b) (c) (d) (e)

Fig. 3. Pseudo-color images and ground truth maps of the five HAD datasets. (a) WHU-Hi-River. (b) Los Angeles-I. (c) Los Angeles-II. (d) Cat Island. (e) Urban.

Meanwhile, we maintain equal weights for potential anomaly regions as for the basic backgrounds, because we expect these regions to have low reconstruction errors at the pixelized resolution, while simultaneously avoiding strengthening their features. Therefore, the reconstruction weight of the pixelized image can be calculated as follows:

$$\mathbf{W}^{(s)} = \mathbf{J} - \mathbf{B}^{(s)}, \quad s \in \{4, 8\}. \tag{14}$$

In the initial stage of training, the weights for each resolution are fixed at 1. After every η epochs, the reconstruction weights $\mathcal{W} = \{\mathbf{W}^{(1)}, \mathbf{W}^{(4)}, \mathbf{W}^{(8)}\}$ are updated according to the above process. For each resolution, the loss function $\mathbf{L}^{(s)}$ can be calculated as follows:

$$\mathbf{L}^{(s)} = \frac{1}{hw} \sum_{i=1}^{h} \sum_{j=1}^{w} \mathbf{W}_{i,j}^{(s)} \left\| \mathbf{X}_{i,j}^{(s)} - \hat{\mathbf{X}}_{i,j}^{(s)} \right\|_2^2. \tag{15}$$

Finally, the total loss $\mathcal{L}_{\text{total}}$ of the multiresolution network is

$$\mathcal{L}_{\text{total}} = \mathbf{L}^{(1)} + \mathbf{L}^{(4)} + \mathbf{L}^{(8)}. \tag{16}$$

IV. EXPERIMENTS

A. Dataset Description

In this work, five real HSIs from different scenes are used to verify the HAD performance of the proposed PixDiff. Their pseudo-color images and ground truths are shown in Fig. 3. The detailed descriptions are as follows.

1) WHU-Hi-River: This dataset was collected by the Nano-Hyperspec visible and near-infrared UAV-borne hyperspectral sensor. The image has a size of 105 × 168 × 135, containing 21 anomaly pixels. The spatial resolution is 6 cm/pixel.

2) Los Angeles-I: This dataset was collected by the AVIRIS sensor, with 7.1 m/pixel as the spatial resolution. It has a size of 100 × 100 × 205, containing 144 anomaly pixels.

3) Los Angeles-II: This dataset was collected by the AVIRIS sensor, with 7.1 m/pixel as the spatial resolution. It has a size of 100 × 100 × 205, containing 170 anomaly pixels.

4) Cat Island: This dataset was collected by the AVIRIS sensor, with 17.2 m/pixel as the spatial resolution. It has a size of 150 × 150 × 188, containing 19 anomaly pixels.

5) Urban: This dataset was collected by the AVIRIS sensor, with 7.1 m/pixel as the spatial resolution. It has a size of 100 × 100 × 205, containing 272 anomaly pixels.

B. Experimental Settings

1) Comparative Methods: We conducted comparative experiments with 12 existing HAD methods to verify the detection performance of the proposed PixDiff. The statistic or representation learning-based methods include RX [9], CRD [21], LRASR [22], 2S-GLRT [20], and PTA [27], while LREN [58], Auto-AD [12], BSDM [40], BockNet [50], GT-HAD [30], DFAN [52], and MSNet [31] are deep learning-based methods. The experimental settings of the comparison methods follow the optimal settings provided in the original papers or the default settings in the official code.

2) Implementation Details: The proposed PixDiff is implemented using the PyTorch framework. To demonstrate the generality of PixDiff, hyperparameters are uniformly set across all datasets. The learning rate is set to 1×10^{-6}, the resolution balance parameter λ is set to 0.2, and the weight update interval η is set to 300. Input images are reduced to 20 bands using Principal Component Analysis (PCA), followed by Z-score normalization. All deep learning experiments are conducted on a single NVIDIA GeForce RTX 2080 Ti with 11 GB of GPU memory.

3) Evaluation Metrics: In this work, we plot the receiver operating characteristic (ROC) curve about the true positive rate P_d, false positive rate P_f, and the corresponding threshold τ of the detection map. Then, the corresponding area under the ROC curve (AUROC) values are calculated. AUROC is the core metric of HAD, with higher values indicating stronger discrimination ability between background and anomalies. To further demonstrate the methods' sensitivity to anomalies, we introduce the area under the precision–recall curve (AUPRC) metric. In addition, referring to the industrial anomaly detection field, we introduce the per-region overlap (PRO) score [59]. We plot the PRO curve within the false positive rate interval of [0, 0.3] and calculate the normalized area under the PRO curve (PRO-AUC), as an approximate average of PRO scores under different false alarm rates. PRO-AUC treats each anomalous region equally regardless of size, reflecting the method's detection completeness for each anomaly target under low false positive rates. The three metrics are combined

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TABLE II
PRO-AUC SCORES FOR DIFFERENT METHODS ON THE FIVE DATASETS

Dataset	RX	CRD	LRASR	2S-GLRT	PTA	LREN	Auto-AD	BSDM	BockNet	GT-HAD	DFAN	MSNet	PixDiff
WHU-Hi-River	0.9960	0.9160	0.7227	0.9690	0.6540	0.9413	0.9957	0.9960	0.9980	0.9927	0.9957	**0.9983**	**0.9983**
Los Angeles-I	0.6763	0.5507	0.2073	0.5313	0.6817	0.5617	0.6547	0.6763	0.8033	0.5480	0.6790	0.7543	**0.7990**
Los Angeles-II	0.7727	0.2267	0.0540	0.2303	0.3990	0.6987	0.7263	0.7727	0.8307	0.7233	0.7760	0.7917	**0.8613**
Cat Island	0.9287	0.9813	0.6483	0.9863	0.9463	0.6933	0.9297	0.9287	0.9460	0.9953	0.9297	0.9960	**0.9993**
Urban	0.9393	0.4513	0.0917	0.0000	0.5007	0.9007	0.9360	0.9393	0.9063	0.9260	0.9410	0.9413	**0.9623**
Mean	0.8626	0.6252	0.3448	0.5434	0.6363	0.7591	0.8485	0.8626	0.8969	0.8371	0.8643	0.8963	**0.9241**

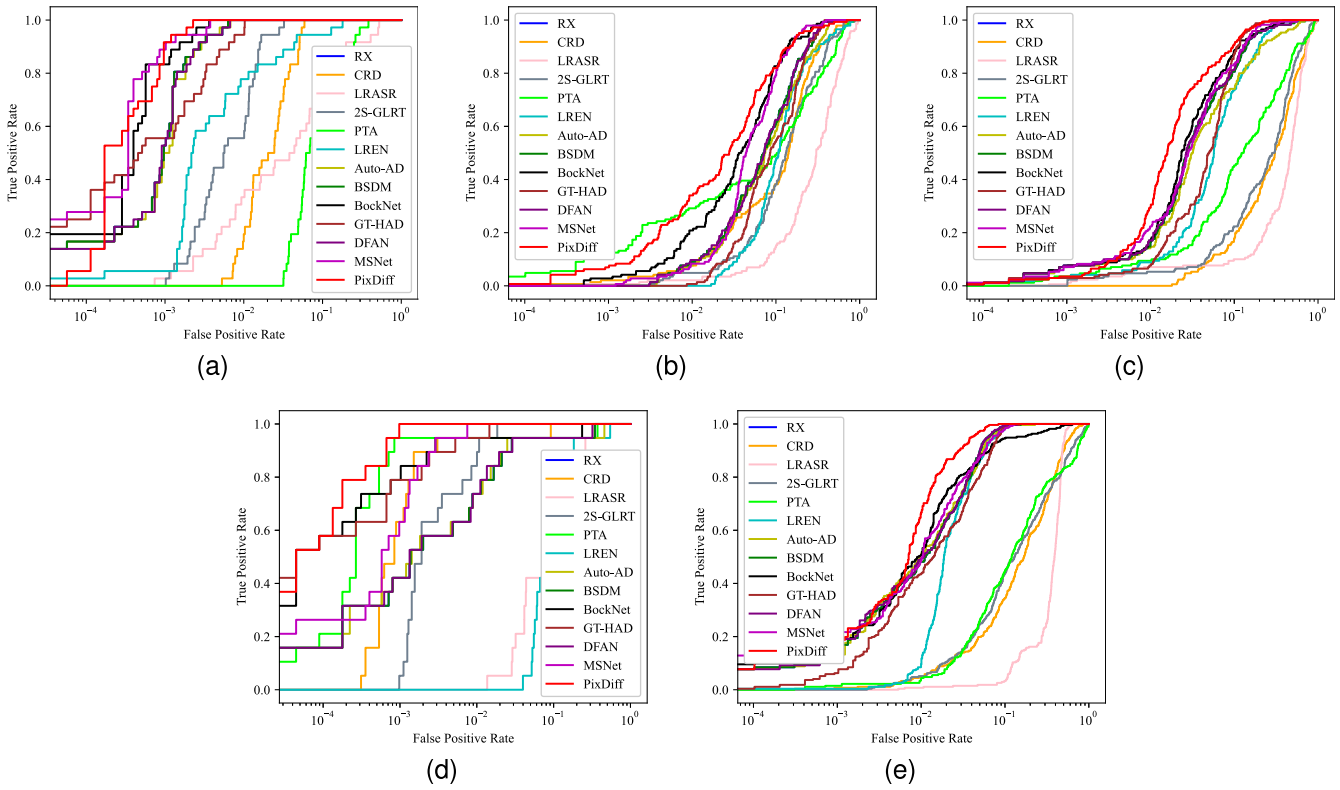


Fig. 4. ROC curves for different methods on the five HAD datasets. (a) WHU-Hi-River. (b) Los Angeles-I. (c) Los Angeles-II. (d) Cat Island. (e) Urban.

to reflect the detection performance and characteristics of a method comprehensively.

C. Comparison Results

We compared the detection performance of the proposed PixDiff with 12 existing methods on five datasets. The AUROC and PRO-AUC scores of each method are shown in Tables I and II, and the ROC curves are displayed in Fig. 4. As can be seen, the proposed PixDiff achieved the best AUROC and PRO-AUC results on all five datasets, which proves that PixDiff not only has good pixel-level detection capability, but also can accurately locate each anomaly target with few omissions. Particularly on Los Angeles-I and Los Angeles-II datasets, PixDiff shows significant performance improvements compared with other methods. From the pseudo-color images, it can be observed that these two datasets have complex background components and high detection difficulty, demonstrating PixDiff's ability to suppress complex backgrounds. The visualization results of detection maps are shown in Fig. 5, showcasing PixDiff's excellent anomaly sensitivity and background suppression abilities. We additionally plotted box plots of background and anomaly distributions in detection results for each method in Fig. 6. The upper and lower bounds of the box represent the 90th and 10th percentiles, with the middle line indicating the median. The whiskers extend to the maximum and minimum values. From the box plots, it can be seen that PixDiff achieves a good balance in background suppression and background-anomaly separation ability.

To further analyze the performance of each method, we calculated the AUPRC scores of the results in Table III. AUPRC comprehensively evaluates the precision and recall of methods, focusing on their performance on anomalies. It can be seen that PixDiff achieved the best AUPRC scores on four datasets except WHU-Hi-River. This may be because the target size in the WHU-Hi-River dataset is relatively small, while there are significant shadows. Therefore, some centers of the detected targets are shifted, leading to an increase in the false positive rate. For the Cat Island dataset, although the AUROC score of PixDiff is close to 2S-GLRT, GT-HAD, and MSNet, its AUPRC score shows a significant improvement. From the visualization results, we can see that PixDiff can

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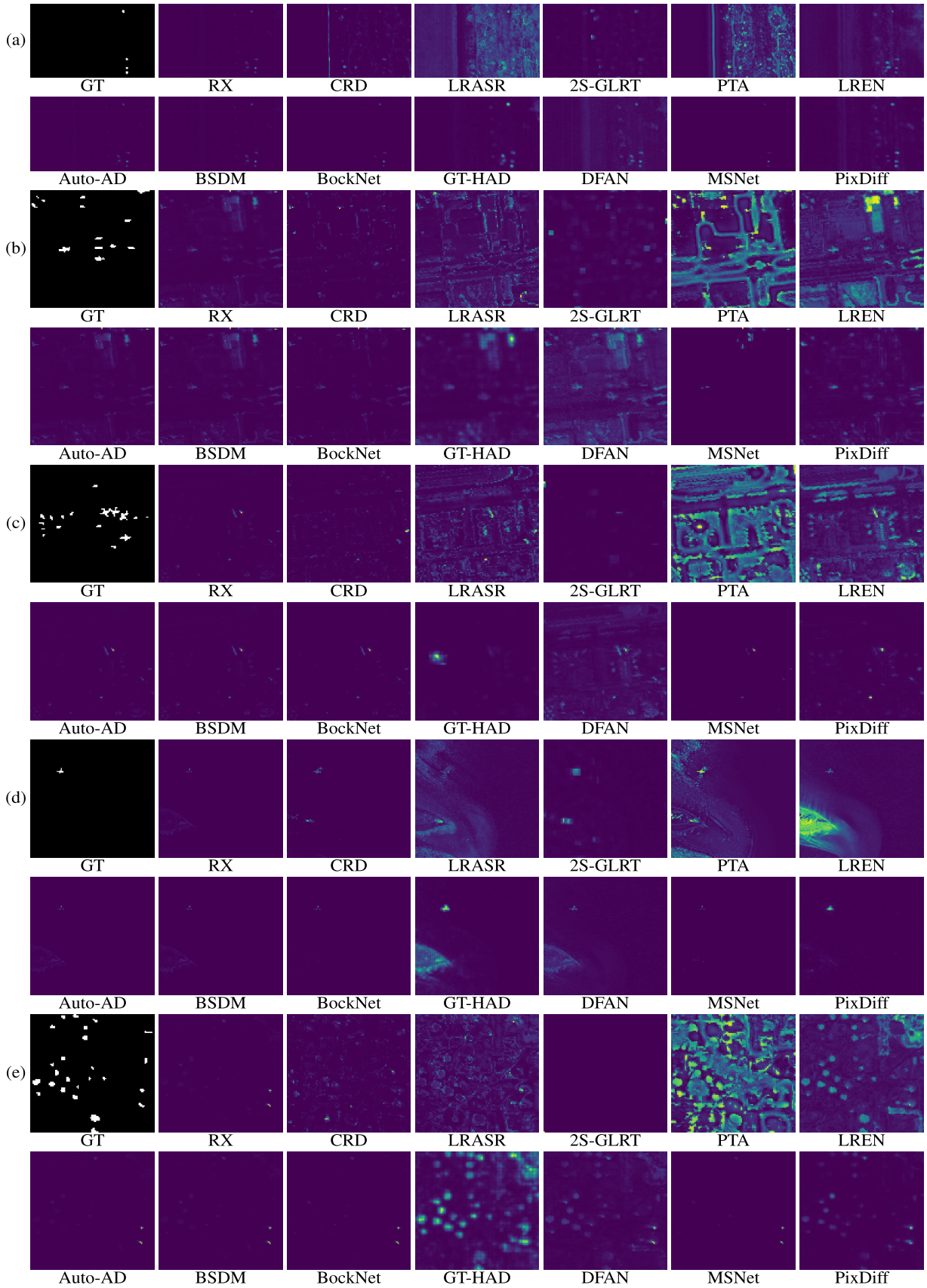


Fig. 5. Anomaly detection maps of different methods on five datasets. (a) WHU-Hi-River. (b) Los Angeles-I. (c) Los Angeles-II. (d) Cat Island. (e) Urban.

clearly detect anomalies while significantly suppressing the complex background in the lower left corner.

Table IV presents a comparison of model parameters, FLOPs, and running time of deep learning-based methods

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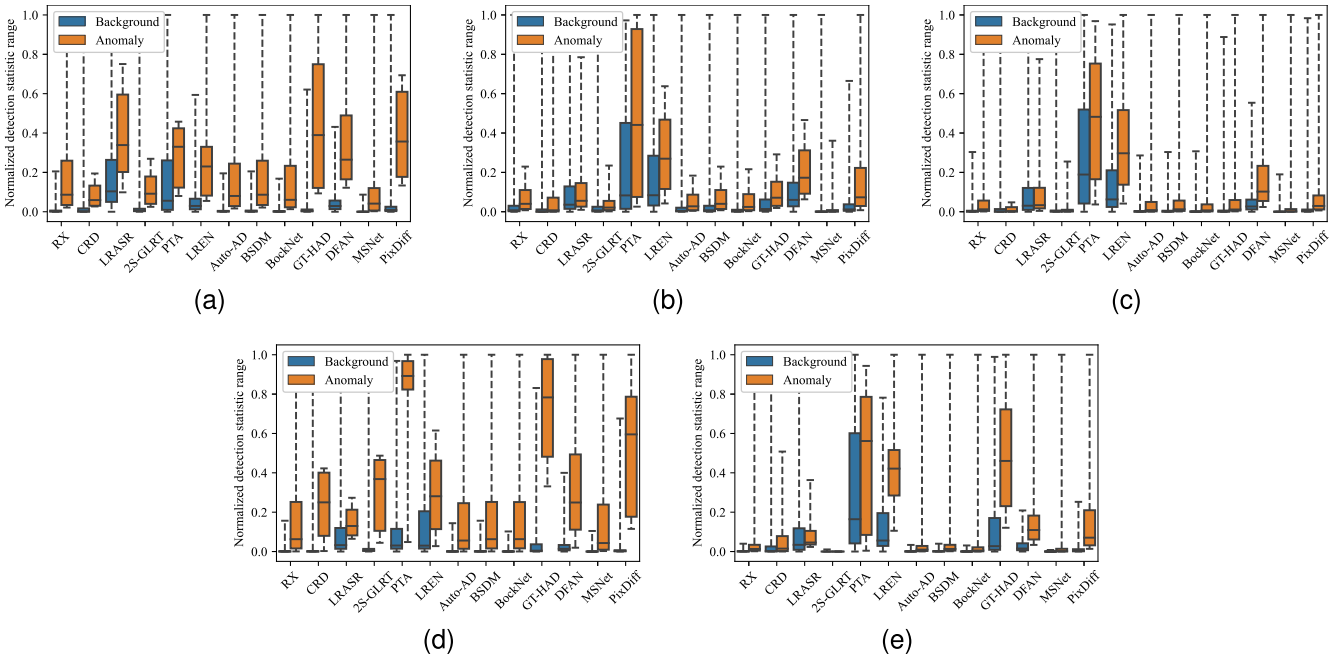


Fig. 6. Box plots for different methods on the five HAD datasets. (a) WHU-Hi-River. (b) Los Angeles-I. (c) Los Angeles-II. (d) Cat Island. (e) Urban.

TABLE III
AUPRC SCORES FOR DIFFERENT METHODS ON THE FIVE DATASETS

Dataset	RX	CRD	LRASR	2S-GLRT	PTA	LREN	Auto-AD	BSDM	BockNet	GT-HAD	DFAN	MSNet	PixDiff
WHU-Hi-River	0.5784	0.0411	0.0401	0.1254	0.0101	0.2256	0.5658	0.5784	0.7349	0.6321	0.5714	**0.7929**	0.7189
Los Angeles-I	0.0866	0.0741	0.0233	0.0471	0.2334	0.0523	0.0798	0.0866	0.1611	0.0637	0.0872	0.1112	**0.2634**
Los Angeles-II	0.2291	0.0264	0.0287	0.0395	0.0831	0.1351	0.2064	0.2291	0.2509	0.1591	0.2339	0.2439	**0.3065**
Cat Island	0.3333	0.2936	0.0049	0.1308	0.6203	0.0047	0.3320	0.3333	0.7302	0.7071	0.3361	0.5018	**0.8385**
Urban	0.5852	0.0790	0.0339	0.0877	0.0954	0.3428	0.5893	0.5852	0.6096	0.4906	0.5905	0.6133	**0.6909**
Mean	0.3625	0.1028	0.0262	0.0861	0.2085	0.1521	0.3547	0.3625	0.4973	0.4105	0.3638	0.4526	**0.5636**

TABLE IV
EFFICIENCY ANALYSIS OF DEEP LEARNING-BASED METHODS ON URBAN DATASET

Method	Params (M)	FLOPs (G)	Time (s)	mAUROC
LREN	0.03	1.81	71.55	0.9234
Auto-AD	2.99	4.98	29.30	0.9497
BSDM	0.92	3.46	18.01	0.9589
BockNet	0.59	10.82	19.11	0.9687
GT-HAD	0.04	34.01	43.46	0.9589
DFAN	**0.01**	**0.02**	**12.50**	0.9598
MSNet	0.02	0.21	34.36	0.9718
PixDiff	39.17	19.70	46.01	**0.9795**
PixDiff-Lite	8.27	4.66	25.22	0.9765

on the Urban dataset. Due to the utilization of the diffusion model and multiresolution structure, PixDiff exhibits higher parameter counts and computational costs compared with other methods. However, it is worth noting that the current implementation of PixDiff focuses on addressing the challenges of anomaly visibility during training and the confusion with complex backgrounds. The network architecture prioritizes detection accuracy over efficiency, leaving potential for efficiency enhancement. To further validate the feasibility of efficiency improvement for PixDiff, we designed a lightweight version, PixDiff-Lite, by reducing the number of channels in each layer of the network. As shown in the table, although PixDiff-Lite still has a relatively high parameter count compared with other methods, its FLOPs and running time have decreased to a moderate level while maintaining the best mean AUROC score. This demonstrates that PixDiff has good potential for efficiency optimization while preserving the detection performance. Therefore, in resource-constrained applications, PixDiff is expected to achieve further efficiency improvements through more compact network design to reduce redundancy, or model compression techniques such as pruning.

In summary, comparison results have demonstrated that the proposed PixDiff has excellent anomaly detection capabilities while effectively suppressing the background. For complex background regions, PixDiff can better separate them from anomalies compared with existing methods, thereby improving the detection performance.

D. Ablation Studies

In this section, we validate the effect of each component in PixDiff. As shown in Table V, starting from the basic U-Net, we gradually add the diffusion denoising process, multiresolution (MR) pixelization, SIM, and multiresolution adaptive weights (AW). The AUROC scores and AUPRC scores on five datasets are listed in the table.

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TABLE V
EFFECT OF MR PIXELIZATION, SIM, AND ADAPTIVE WEIGHTS ON THE DETECTION PERFORMANCE ACROSS FIVE DATASETS

Framework	MR	SIM	AW	WHU-Hi-River		Los Angeles-I		Los Angeles-II		Cat Island		Urban	
				AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
U-Net				0.9989	0.5965	0.9248	0.1172	0.9492	0.2507	0.9914	0.3362	0.9808	0.5984
DDPM				0.9988	0.6055	0.9313	0.1372	0.9531	0.2715	0.9914	0.3612	0.9832	0.6220
	✓			0.9989	0.6081	0.9322	0.1361	0.9549	0.2636	0.9916	0.3735	0.9838	0.6369
	✓	✓		0.9994	0.6783	0.9350	0.2288	0.9548	0.2383	0.9996	0.8031	0.9874	0.6687
	✓	✓	✓	**0.9995**	**0.7189**	**0.9418**	**0.2634**	**0.9672**	**0.3065**	**0.9998**	**0.8385**	**0.9884**	**0.6909**

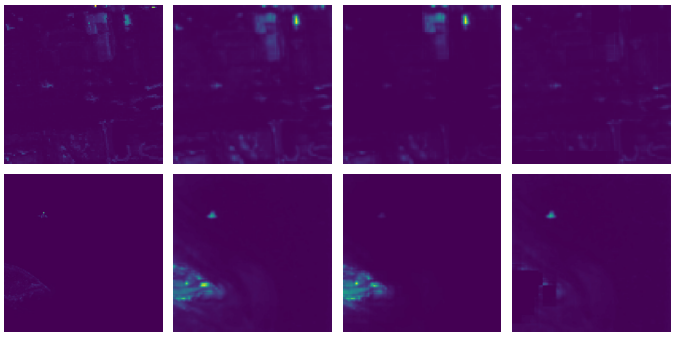


Fig. 7. Visualization of the effect of SIM on Los Angeles-I and Cat Island datasets. (a) Detection result of BSDM. (b) Original reconstruction error of PixDiff. (c) Extracted complex background by SIM. (d) Potential anomaly map of PixDiff.

From the table, it can be seen that the introduction of the diffusion denoising process has generally improved the detection performance. Through gradually learning the reconstruction process from noise to image, the network is more inclined to reconstruct backgrounds that approximate Gaussian noise. The multiresolution pixelization alone has a marginal improvement in the detection performance. However, it establishes a critical foundation for the functionality of the other components. Through the reconstruction by multiresolution branches, SIM is able to analyze the differences between anomalies and complex backgrounds across resolutions, thereby effectively separating anomalies from complex backgrounds. Multiresolution adaptive weights utilize the separation results of SIM to guide the model's learning approach for anomalies and complex backgrounds, respectively. Therefore, these components need to collaborate to achieve the optimal detection performance, rather than exhibiting a simple additive relationship. After introducing the SIM, the overall detection performance of the network has improved significantly. Especially for the Cat Island dataset, the AUPRC score increased from 0.3735 to 0.8031. Based on the complex backgrounds and potential anomalies extracted by SIM, after incorporating multiresolution adaptive weights during training, the detection performance is further improved. The network can learn enhanced reconstruction of complex backgrounds and ignore anomalies, thereby generating reconstructed images that are more similar to the background.

To further demonstrate the separation performance of complex backgrounds and anomalies, we show the visualized extraction results in Fig. 7. In another diffusion denoising-based method BSDM, the background is treated as a type of noise and gradually suppressed through the denoising process. However, BSDM does not specifically process complex background details and only suppresses the background based on overall distribution characteristics. Consequently, it may simultaneously highlight or suppress both complex backgrounds and anomalies. As shown in Fig. 7(a), the detection results of BSDM exhibit similar response levels for both anomalies and complex backgrounds. In contrast, PixDiff can effectively separate complex backgrounds and anomalies through SIM. In the original reconstruction error, besides anomalies, there are also significant complex background regions. In the extraction results of the SIM, the main parts of the complex background are extracted. In the final obtained potential anomaly maps, the complex background is significantly suppressed while anomalies are preserved. The SIM achieves good separation between complex background and anomalies, thus having a powerful effect on improving the network's detection performance.

E. Analysis of Multiresolution Configuration

In this work, we select the original resolution, 4×4 pixelization, and 8×8 pixelization as the multiresolution configuration. To validate the rationality of this setting, we compare the effect of different combinations of various pixelization block sizes on the detection performance. Since the size of anomalies is unknown in practical applications, it is difficult to customize appropriate pixelization block sizes for each dataset. Therefore, we need to find a universal size combination within a reasonable range. For convenient selection of appropriate size combinations, we select 2×2, 4×4, 8×8, and 16×16 as representatives of different scale levels. The detection results of various combinations are shown in Table VI.

From the two-branch experiments, it can be observed that using 2×2 or 16×16 pixelization alone may lead to significant performance degradation on some datasets. In the multibranch experiments, we find that incorporating 2×2 pixelization usually results in decreased detection accuracy. This may be because excessively small sizes fail to destroy anomaly structures, thus are unable to reduce anomaly visibility and distinguish them from complex backgrounds. Although 16×16 pixelization performs relatively poorly on most datasets, it appears to improve the detection performance on the Urban dataset. This may be attributed to the larger-sized anomalies and fewer significant complex backgrounds in the Urban dataset, allowing larger pixelization sizes to better destroy anomaly structures and thereby enhance the detection performance.

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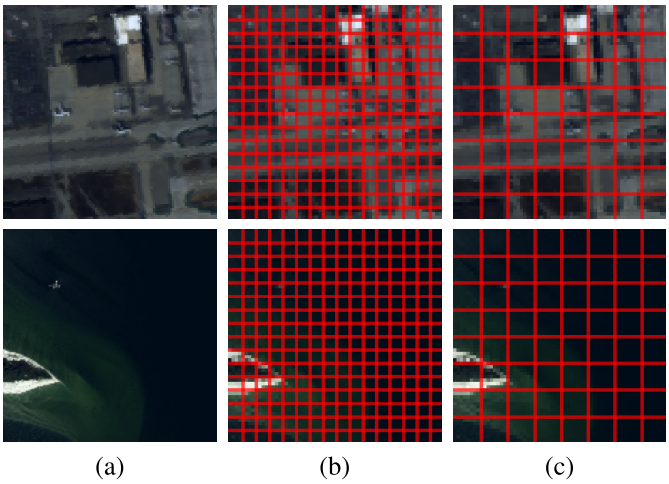
TABLE VI

AUROC SCORES FOR DIFFERENT PIXELIZATION BLOCK SIZE COMBINATIONS ON THE FIVE DATASETS. RIVER REPRESENTS THE WHU-HI-RIVER DATASET, LA-I REPRESENTS LOS ANGELES-I DATASET, LA-II REPRESENTS LOS ANGELES-II DATASET, AND CI REPRESENTS THE CAT ISLAND DATASET

Size of Pixelization Block					River	LA-I	LA-II	CI	Urban	Mean
1×1	2×2	4×4	8×8	16×16						
✓	✓				0.9992	0.9415	0.9256	0.9993	0.9777	0.9687
✓		✓			0.9992	0.9408	0.9612	0.9997	0.9863	0.9774
✓			✓		0.9994	0.9433	0.9555	**0.9998**	0.9838	0.9764
✓				✓	0.9815	0.9276	0.9351	**0.9998**	0.9867	0.9661
✓	✓	✓			0.9988	0.9400	0.8965	0.9997	0.9887	0.9647
✓		✓	✓		**0.9995**	0.9418	**0.9672**	**0.9998**	0.9891	**0.9795**
✓		✓		✓	0.9986	0.9377	0.9443	**0.9998**	0.9894	0.9740
✓			✓	✓	0.9984	**0.9473**	0.9455	**0.9998**	0.9888	0.9760
✓	✓	✓	✓		0.9993	0.9205	0.9476	**0.9998**	0.9895	0.9713
✓	✓		✓	✓	0.9990	0.9341	0.9492	**0.9998**	**0.9909**	0.9746

Furthermore, taking Los Angeles-I and Cat Island datasets as examples, we demonstrate the relationship between pixelization size and the sizes of HSI contents. As shown in Fig. 8, for Los Angeles-I dataset, 8 × 8 pixelization can effectively cover anomalies, ensuring that each anomaly does not occupy an excessive proportion within a block, while preserving complex backgrounds such as the building in the upper region. Table VI also shows that for Los Angeles-I dataset, the 8 × 8 pixelization branch contributes most significantly to the improvement of the detection performance. For the Cat Island dataset, both 4 × 4 and 8 × 8 pixelization are sufficient to destroy anomaly targets while preserving the complex background region below, resulting in similar AUROC scores.

To verify the generalizability of PixDiff under unified parameter settings, we select 4 × 4 and 8 × 8 pixelization branches for all datasets. In practical applications, the pixelization size can be adjusted according to the expected size of anomalies.



(a) (b) (c)

Fig. 8. Visualization of pixelization block sizes. (a) Original HSI. (b) Pixelization with 4 × 4 blocks. (c) Pixelization with 8 × 8 blocks. The red grids illustrate the pixelization block boundaries.

F. Parameter Analysis

In this section, we analyze the effect of the resolution balance parameter λ, the weight update interval η, and the number of principal components through PCA on the detection performance. When experimenting with one parameter, the remaining experimental settings are fixed. The results are shown in Fig. 9.

The resolution balance parameter λ controls the influence of pixelized images at different resolutions in SIM. As shown in Fig. 9(a), the value of λ has little impact on detection results of four datasets. Only for Los Angeles-II dataset, when λ exceeds 0.7, the detection performance begins to decrease significantly. This might be because the anomaly regions in Los Angeles-II dataset are large and compact, making it difficult for small-scale pixelization to destroy the structures of anomalies. On the other hand, when λ is small, there is a gradual increase in the detection performance on the Urban dataset. This could be because the background of the Urban dataset contains some detailed complex areas that are more difficult to reconstruct in small-scale pixelized images, thus enabling extraction.

The weight update interval η controls the update frequency of adaptive weights. As shown in Fig. 9(b), for WHU-Hi-River, Cat Island, and Urban datasets, the value of η does not significantly affect the detection performance. For the other two datasets, the detection performance fluctuates within a certain range. This may be because Los Angeles-I and Los Angeles-II datasets have more complex content, resulting in larger variations in reconstruction errors and weights during the initial network training. Although there is some uncertainty, the detection performance does not exhibit a significant decline, and the model remains generally stable under different parameter settings.

In addition, we analyzed the impact of selecting different numbers of principal components through PCA dimensionality reduction on the detection performance. As shown in Fig. 9(c), when the number of principal components is between 10 and 30, the detection performances show advantages and disadvantages on different datasets. However, when the number of principal components continues to increase, the detection performance on Los Angeles-I and Los Angeles-II datasets shows a significant decline. This may be because as the number of selected principal components increases, the information in later bands decreases significantly while noise increases. Since diffusion denoising models recover images from noise, bands with low signal-to-noise ratios may affect the model's ability to recover meaningful information, especially for these two

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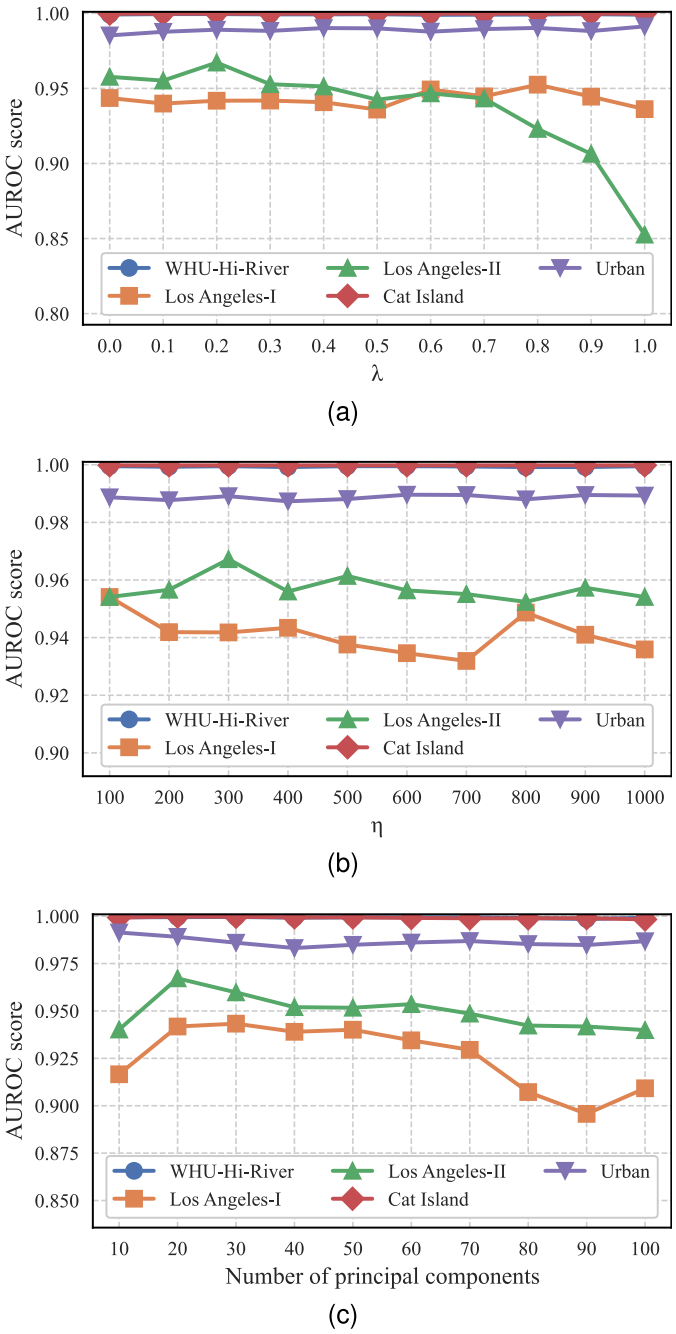


Fig. 9. AUROC score of PixDiff on different parameter settings. (a) Resolution balance parameter λ. (b) Weight update interval η. (c) Number of principal components through PCA.

relatively complex datasets. Therefore, selecting around 20 principal components can better preserve image information while reducing noise introduction.

V. Conclusion

In this article, we propose an HAD method called PixDiff based on a multiresolution diffusion denoising model. The pixelization operation is innovatively introduced into HAD, thereby destroying potential anomaly structures. Subsequently, an SIM is proposed, which effectively extracts and suppresses complex backgrounds in the detection map by analyzing reconstruction tendencies across different resolutions. Furthermore, a multiresolution adaptive weighted reconstruction loss is designed to guide the network in learning different treatments for complex backgrounds and anomalies at different resolutions. Experimental results demonstrate that the proposed PixDiff achieves the superior detection performance on the five datasets. Our research provides a novel perspective on distinguishing between anomalies and backgrounds in unsupervised HAD. In future work, we will focus on enhancing the generalizability of HAD models, such as adaptively learning the structural transform scales, to enable their application in more diverse and unknown scenarios.

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